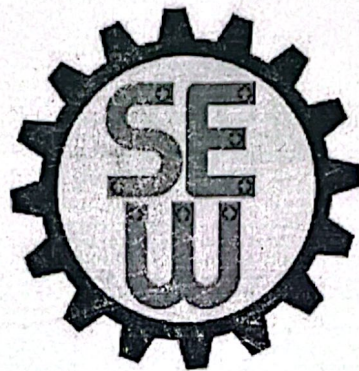


Toffee 400 kg/hr.

BOOKLET FOR COMPLETE PRODUCT LINES



Starmers
Engineering Works

Eclair Toffee (400 KG/HOUR)

- 1: Glucose Tank ✓
- 2: Steam Generator (600 kg) ✓
- 3: Steam Headler ✓
- 4: Toffee Cooker (150 kg) ✓
- 5: Cooling Table ✓
- 6: Pulling Machine ✓
- 7: Liquid Filling Pump ✓
- 8: Batch Roller (75kg) ✓
- 9: Rope Sizer (5 stages) ✓
- 10: Uniplast (160-A) ✓
- 11: Rotary Die ✓
- 12: Die Stand ✓
- 13: Ac Unit ✓
- 14: Cooling Tunnel 3 Stages ✓
- 15: Wrapping Machine ✓

MH
Approved

1. Glucose Tank SEW-101

Trait:

Glucose storage tanks are engineered with specific traits to ensure hygienic, stable and crystal free storage of liquid glucose and Syrups for industrial applications.

Specifications:

- * Grade: S.S 304
- * Sheet thickness: 5 mm upper and lower sheet
- * Outer dia sheet thickness: 3 mm
- * Design Type: Vertical
- * Heating System: Coil on the base of tank for Steam Circulation
- * Ladder: Outside the tank
- * Capacity: 20 ton Glucose

- * Dimension: 16ft height x 7ft diameter
- * Power load: 0

Approved

2. Steam Generator SEW-102

Trait:

Steam generator heats water to produce high Pressure steam for heating, cleaning or industrial processes.

Detail of Mechanical Parts:

1. Gauge Glass: 01 ✓
2. Stop Valve diameter 1": 01 ✓
3. Safety Valve diameter 1": 01 ✓
4. NRV diameter 1": 01 ✓
5. Blow down diameter 1": 01 ✓
6. Air Vent diameter globe Valve = 1/2": 01 ✓
7. Feed Water Pump 01 Multistage (Grundfos): 01 ✓
8. Electric panel Box with complete auto controlled system: 01 ✓
9. Stack diameter 8" height 15ft. thick 2.5 mm: 01 ✓

Detail of Electric Parts:

1. Burner with gas assembly and diesel pump ✓
2. Panel Box 16" x 30" x 8" thick 18g ✓
3. Overload company Schneider: 02 ✓
4. Pressure switch 10 kg imported: 02 ✓
5. Electric Contactor: 02 ✓
6. Power relay company (Finder): 04 7. Push button company (Telemecanique): 03 ✓

Specifications:

- * Type of passes: 3 pass ✓
- * Test pressure: 225 psi ✓
- * Working pressure: 150 psi ✓
- * Tube Dia: 2" or 2 1/2" ✓
- * Pressure Shell thickness: 1/2" ✓
- * Tube sheet thick: 1/2" ✓
- * Rock wool thickness: 2"
- * Isolation sheet thickness: 24 GAUGE

Approved

* Burner Motor: 2hp ✓

* Feed water motor: 3hp ✓

General Details: ✓

* Capacity: 600 kg/hr ✓

* Electric Load: 5 hp ✓

* Dimension: 8' x 3'5" x 3'5" ✓

Approved by
Arsalan Ghahed

3. Steam Header SEW-103

Trait:

Steam headers are design to collect steam from a boiler or steam generator and distribute highquality, dry steam to various user points while performing several critical functions. ✓

Specifications:

- * Material of header: M.S ✓
- * Diameter: 1 ft ✓
- * Branch pipes: 1", 2", 3", 4" depend on system needs ✓
- * Design: Cushioned Cylinder ✓
- * Connection: 5 ✓
- * Dimension: Dia 1ft x Length 3ft ✓
- * Power load: 0 ✓


Approved

4: Toffee Cooker (150 Kg)

Trait:

The purpose of a toffee cooker is to heat and cook sugar based mixtures to a controlled temperature so the toffee reaches the correct thickness, texture, and moisture level. It provides uniform heating and precise temperature control to prevent burning, ensure consistent quality, and achieve the required hardness or chewiness of the final toffee product.

Specifications:

Toffee cooker

- * Capacity: 150kg per batch after 20 to 25 mins
- * Stirrer: S.S 304
- * Pan material: S.S 304
- * Pan double jacketed with S.S 304 / M.S optional
- * Frame in: M.S / S.S 201 optional
- * Electric pannel with all new features
- * 3 hp gear motor drive
- * Safety valve
- * Cover of pan in S.S 304

Approved

Dimension (H/W/D) ✓

POWER LOAD = 3hp ✓

5. Cooling Table SEW-108

Trait:

The cooling table is a crucial post-cooking unit used for rapid cooling and conditioning of the cooked sugar mass before further processing such as mixing, flavoring, coloring, or forming. It ensures uniform texture and workability of the batch.

Specifications

- * Base plate thickness: 10mm in S.S 304
- * Material of plate: Boundary wall S.S 304
- * Cooling medium water: Cooling system A frame work for moving water Inlet and outlet Valve.
- Legs: 4 Nos in S.S 201 grade

Dimension: 3ft x 6ft x 2'6" (W x L x H)

Power Load: 0

18/4/24
Approved

6: Pulling Machine

Trait:

The purpose of a pulling machine is to stretch and fold cooked candy or toffee in a controlled way to improve its texture and structure. By continuously pulling the mass, the machine incorporates air, evens out temperature, and develops the desired smoothness, shine, and chew of the final product.

Specifications:

- * Capacity: 100kg ✓
- * Main drive motor: 10hp ✓
- * Arms: Hard chromed ✓
- * Arms for continuous pulling: 3 arms ✓
- * Side sheets: In S.S 201 ✓
- * Electric pannel for control ✓

~~Not~~
Approved

Dimension (H/W/L) ✓

POWERLOAD = 10hp ✓

7: Liquid Filling Pump

Trait:

The center filling pump is a precision dosing system designed to inject liquid filling (such as fruit syrup, chocolate, cream, or jam) into the core of the candy rope during the forming process. It ensures consistent filling quantity and uniform distribution across all candies.

Specifications:

- * Filling chocolate and liquid ✓
- * keef in S.S 304 ✓
- * keef double-jacketed ✓
- * Rod in S.S with Teflon covering food-grade heat proof material ✓
- * Pipe flexible food grade ✓
- * Electric panel with new feature ✓
- * Gear motor 2 hp main drive ✓
- * Stand in M.S ✓
- * Electric heater to moderate temperature 1000 watt 1 NOS ✓
- * Electric inverter 2HP to increase or decrease filling in product ✓
- * Overflow system to recycle liquid/Chocolate ✓

Dimension: (H \ W \ L) ✓

Power Load: 3.34 hp ✓

*12/11/24
Approved*

8: Batch Roller (75kg)

Trait:

The batch roller is a key intermediate machine in the candy, lollipop, toffee, and fudge production line. It is designed to continuously roll, shape, and form the cooked mass into a uniform cylindrical rope. It ensures consistent rope size and proper distribution of center filling, flavor, and color before feeding into the rope sizer or forming machine.

Specifications:

- * Frame work horizontal
- * Rollers Quantity 4
- * Rollers material in S.S 304 food grade
- * Upper body in S.S 304 food grade
- * Bearing in gears
- * Gear motor main drive 1hp
- * Up and down drive 1hp
- * Electric heaters to maintain temperature 1000 watt (2 NOS)
- * Capacity 75 kg
- * Length 6.5 ft
- * Gear motor SEW brand Germany
- * Roller caplene in casting material
- * Additional tray for removal of wastage

Dimensions (H x W x L): 4'6" x 2'2" x 7'

Power Load: 4.68 hp

Approved

9: ROPE SIZER SEW-212

Traits:

The rope sizer is a precision machine used to accurately reduce & calibrate the diameter of the candy and toffee rope produced by the batch roller. It ensures a uniform rope thickness suitable for feeding into the forming or cutting machine, maintaining consistent candy size and weight throughout production.

Specifications:

- * Horizontal model ✓
- * 5 stages (4 stages horizontal & 1 vertical) ✓
- * 1st pair in S.S 304 37mm ✓
- * 2nd pair in S.S 304 32mm ✓
- * 3rd pair in S.S 304 27mm ✓
- * 4th pair in S.S 304 22mm ✓
- * 5th pair in S.S 304 17mm ✓
- * Electric panel in S.S 201 ✓
- * Sizer plates in S.S 304 ✓
- * Side Sheets S.S 201 ✓
- * Rope guide in brass ✓
- * Electric inverter new brand INVENT company 3HP ✓
- * Gear motors are of SEW brand Germany 3hp ✓
- * Heat controllers 2 NOS ✓
- * Emergency break to stop equipment ✓
- * Emergency switch to off equipment ✓
- * Electric breakers *brand new* ✓
- * Electric components brand new ✓
- * Electric heaters 500 x 4 = 2KW ✓

Approved

Dimension: (H x W x L) 4'6" x 3' x 5'6" ✓

Power Load: 5.68hp ✓

Trait:

10: Uniplast (160-A)

Trait:

- * The SEW-4 has been designed for the continuous production of seamless sweets with or without center filling using a standard die arrangement. ✓
- * The machine can be used as a part of a complete production plant supplied by Starmer Engineering Works, or integrated into an existing line. ✓
- * Suitable for forming hard boiled candies, center liquid filled candies, center powder filled candies, toffees, chewy-toffee, chewy candies, chewing-gum & bubble gum products. ✓
- * It forms seamless sweets with high depth thickness. ✓

Specifications:

- * 5 hp motor for main drive ✓
- * 5 hp inverter new brand (INVENT Company) ✓
- * Up to ⁴⁰⁰~~450~~ Kg/hr production ✓
- * Electric panel with all new features ✓
- * CAMS used in this machine in bearing steel grade ✓
- * Reverse Cams in S.G iron ✓
- * Gears used in this machine are of high carbon steel forged material ✓
- * ^{upto 2700}~~2500 to 3500~~ pieces/min depending upon size of product ✓
- * Filling % of the candy upto 12% ✓

Approved

Dimension (H x W x L): 6.5ft x 4.5ft x 3.5ft Power Load: 5hp ✓

Trait:

11 : Rotary Die

Trait:

The heart of any candy production line is the rotary dies in the forming machine. They literally shape the product and add to its face value. The forming dies are used for the production of hard-boiled candies, Toffees, Eclairs, Powder-filled candies, liquid-filled candies, and more. Starmer Forming dies are recognized in the confectionery industry for their precision-based performance, safety, and functional efficiency.

Starmer machines are built for generational use, we offer a supply of spare parts and die-heads for models that may have been discontinued for decades. As the leading confectionery machinery manufacturer, Starmer aims to be your reliable long-term business partner, and offers the following benefits:

Accuracy:

- * Perfectly formed, seamless candies
- * Nearly any shape of candy can be made
- * To ensure accuracy, we provide drawings, and product samples on request to help you visualize the sense of the product

Quality:

- * The finest quality raw material used
- * Our 55+ years of experience have garnered the forming dies into a piece of perfection, even with continuous production, the die-set does not damage easily
- * Cutting Ring: Cutting Ring in grade Forged M.S. This grade provides cutter pitch (teeth) of 2-3 mm.
- * Punch: Punch in grade 1045, Hardness upto 45 Hrc
- * Block: High Quality
- * M.S. Guide Pins: Guide pins in grade Carbon Steel.
- * Die Parts: Casting in grade Cast Iron.
- * Outer Ring: Outer Ring in Brass Casting. For the improvement of rope cutting.

Durability:

Material specification:

Dimension: [H x W x L] 1ft x 1ft x 1'5"

Power Load: [0]

Approved

Trait:

12: Die Stand

Trait:

It's main purpose is to make die changing and fitting easier and more efficient. ✓

Specification:

- * Main Frame in M.S ✓
- * Wheels for movement ✓
- * Die Caplone in casting Material ✓
- * Screw for die up/down ✓

Dimension:

[H x L]: [5.5ft x 5ft] ✓

Power Load: [0] ✓

Approved

13: 12 Ton AC UNIT

Description

Indoor Unit capacity 6 Ton (7.5 HP) x 2 roof type for cooling conveyor

Accessories Indoor:

1. evaporator brand new copper tube: size 3/8 gage 22 distributor size 1/2 section header 7/8.
2. aluminum fins thickness: 0.150mm company Breez.
3. high velocity Air blower: With 1 HP motor 3 phase 440/v Hz 50.
4. expansion valve: company Emerson model TCLE 7.5 HP.
5. Base frame angle iron sizes: are on customer demand.
6. Indoor frame cover sheet: GI colored 22 gage.
7. Indoor frame cover sheet insulation: with insulated material.
8. Indoor copper pipe fitting: 18 gage length tube.

6 Ton (7.5 HP) x 2 OUTDOOR with all his ACCESSORIES:

1. condensing unit box type.
2. compressor: Copeland scroll capacity 6 Ton 3 phase 440/v Hz 50.
3. Liquid receiver capacity: 5 liter.
4. Oil spreader: in/out pipe size 7/8.
5. Hand valve: 1/2.
6. Filter: 1/2.
7. section accumulator: 7/8.
8. dual high low pressure safety switch: company Danfoss with high pressure leads.
9. Outdoor base frame: channel and angle iron.
10. refrigerant: R-22.

ELECTRIC PANEL equipment's.

1. Magnetic contactor: 30 A for compressor.
2. Holding contactor.
3. Blower contactor.

4. Digital thermostat.
5. Fan contactor.
6. Connection strip.
7. Push button.
8. Electric panel box: 12" x 24".

14: Cooling Tunnel 3 Stages

Trait:

Its main purpose is to rapidly and uniformly reduce the temperature of freshly formed candy, ensuring dimensional stability, proper hardness, and surface quality before cutting, wrapping or packaging. ✓

Specifications:

- * 16ft long ✓
- * 3ft width ✓
- * ~~With 5 ton AC~~ ✓ with 12 ton AC
- * 3 stages ✓
- * 1st step mesh wire in S.S 304 grade ✓
- * 2nd step mesh wire in S.S 304 grade ✓
- * 3rd step mesh wire in S.S 304 grade ✓
- * Outer and inner Side sheets in S.S 201 ✓
- * Electric panel with all new features ✓
- * Inverter new brand (~~Invert 5hp~~) (INVENT 5hp) ✓
- * All covers in S.S 201 ✓
- * Double jacketed insulates with thermopore ✓
- * Rods are in S.S 304 5mm thickness ✓
- * Chain in M.S 40 no ✓
- * Gear motor 3hp (1hp each 3 nos) SEW brand ✓

Approved

Dimension: {H \ W \ L} (x3ftx16ft) ✓

Power Load: 3hp ✓

15: Wrapping Machine: ✓

Trait: ✓

This double twist wrapping machine is built to wrap individual candies quickly, evenly, and without waste. Its main purpose is to take loose candies, feed them one by one, and wrap each piece tightly using single or double paper, keeping the product clean and well protected. The machine also aims to keep production steady and reliable by controlling paper feed, speed, and alignment automatically, so every wrapped candy looks the same and no empty or missing wraps are produced. ✓

MACHINE FEATURES: ✓

1. MACHINE WITH CANDY-SETTLING PLATE, FEEDING CANDY AUTOMATICALLY. ✓
2. SINGLE AND DOUBLE LAYERS OF WRAPPING PAPER ARE USED. ✓
3. 100% PACKING RATE. NO EMPTY PACKAGE. NO CANDY NO PAPER SENDING. ✓
4. SERVO MOTOR CONTROL PAPER SENDING SYSTEM. ✓
5. AUTOMATICALLY FOLLOWED BY PHOTO ELECTRICITY OF COMPUTER. ✓
6. THE SPEED IS ADJUSTED BY FREQUENCY-CHANGING DEVICE. ✓
7. HAS TOUCH SCREEN AND EASY OPERATION. ✓

Specifications :

- * CAPACITY: 450PCS/ MIN. ^{upto} ✓
- * PACKING MATERIAL: CELLOPHANE, PVC, POLYPROPYLENE, STENCIL, COMPOSITE MATERIAL, ETC. ✓
- * POWER: 2.5 KW ✓
- * ELECTRICAL SOURCE: 380V 50HZ ✓
- * WEIGHT OF MACHINE: 1083 KG ✓
- * SPECIFICATION OF FOOD: ✓
- * LENGTH: 11-35MM ✓
- * WIDTH: 11-26MM ✓
- * THICKNESS: 5-20MM ✓
- * DIMENSION: 1900*1100*1480MM ✓

Approved